

TREES AND THEIR DEGREES

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1. INTRODUCTION

In the course of our study of the family of Higman–Thompson groups, there came to be a question that was very easy to answer in small dimensions and much harder in higher dimensions. The fundamental reason for this was that it was easy to come up with a reasonable visualization in low dimensions, and much harder in higher dimensions because of the limitations of 3-dimensional space. Therefore, we tried to come up with a good visualization in higher dimensions.

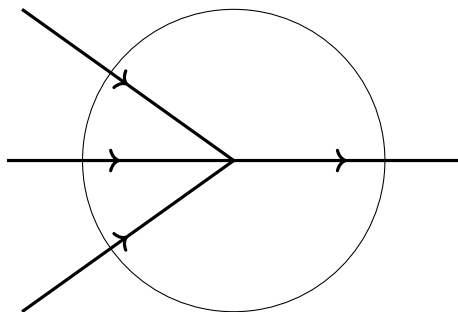
The following document explains this problem and the framework we came up with to answer it in higher dimensions. For extra context on the Thompson groups and their homology, please consult the other file in this folder. Reading the following document is however completely necessary to understanding the specific question we were working on as well as its implementation.

2. TREES

We will consider finite rooted trees, and we call an ***a*-junction** a vertex of the

tree with one parent and a children: For instance,  is a 3-junction.

2.1. Trees with simple junctions. Let us start by considering the following problem. Consider a finite rooted tree which only has a -junctions, for a an integer. We will think of trees as being oriented from the leaves to the root. Consider the following picture:



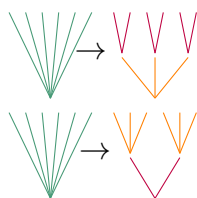
We will call the **degree** of a junction the number of incoming edges minus the number of outgoing edges in a small circle around the junction. For any a -junction, this number is clearly $a - 1$.

Now let us assume we have trees with a -junctions and b -junctions. The a -junctions have degree $a - 1$, and the b -junctions have degree $b - 1$.

2.2. Trees with double junctions. Now let us consider a family of trees with both a -junctions and b -junctions, for a and b two different integers, with the following additional **interchange relation**:



Each a -junction has degree $a - 1$ and each b -junction has degree $b - 1$ again, but what happens when the junctions are switching? We add an ab -junction which represents the switch:



Now we've added an extra junction, and we would like to determine its degree. We can't simply use the same rule that we did before. **Say more about this + put my diagrams with the cubes that come together.**